



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

BT2306-1 – TISSUE ENGINEERING
 (For BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which key factor must be considered when designing scaffolds for tissue engineering?
 A) High porosity and interconnectivity B) High electrical conductivity
 C) Low thermal resistance D) Low water absorption
2. Which fabrication technique is widely used to create porous scaffolds for tissue engineering?
 A) Spectrophotometry B) Gel electrophoresis
 C) Electrospinning D) Atomic force microscopy
3. Which of the following describes the ability of a cell to move towards a chemical signal?
 A) Phagocytosis B) Apoptosis
 C) Endocytosis D) Chemotaxis
4. Which cytoskeletal component is primarily responsible for regulating cell migration?
 A) Microtubules B) Actin filaments
 C) Ribosomes D) Mitochondria
5. Which of the following biomaterials can act as a scaffold for tissue regeneration?
 A) Fibroblast B) Collagen
 C) Osteoclast D) Copper
6. Why is surface modification of biomaterials important for scaffold applications?
 A) To increase thermal resistance B) To improve magnetic properties
 C) To make the material electrically conductive D) To enhance cell attachment and biocompatibility
7. Which feature of biomaterial scaffolds is essential for nutrient exchange in tissue engineering?
 A) Low surface roughness B) High density
 C) Porosity and interconnectivity D) High elasticity
8. How does ECM contribute to the functional integration of implanted materials?
 A) By providing biochemical cues for cell adhesion and growth B) By preventing vascularization of the implant
 C) By increasing immune rejection of the implant D) By forming a barrier between the implant and the host tissue
9. The purpose of growth factor delivery in tissue engineering is to:
 A) To control cellular responses such as differentiation and proliferation B) To increase the degradation rate of biomaterials
 C) To block ECM interactions D) To inhibit tissue regeneration
10. What is the main advantage of polymeric scaffold-based drug delivery systems?
 A) They eliminate the need for controlled drug release B) They provide sustained and localized release of drugs
 C) They degrade within minutes of implantation D) They do not interact with cells or tissues

11. Which material is commonly used in hydrogel-based drug delivery systems?
 - A) Iron oxide
 - B) Alginate
 - C) Polyvinyl chloride (PVC)
 - D) Titanium
12. Polymer microspheres enhance drug delivery:
 - A) By allowing immediate drug degradation
 - B) By providing controlled and targeted drug release
 - C) By forming a rigid scaffold structure
 - D) By preventing cellular uptake of the drug
13. The role of gradient structures in tissue engineering is:
 - A) They mimic natural tissue transitions and promote functional integration
 - B) They ensure uniform mechanical properties throughout the tissue
 - C) They prevent the interaction of different cell types
 - D) They make scaffolds more rigid
14. Which among the following is best for controlled drug delivery?
 - A) Titanium
 - B) Surface modified steel
 - C) Electrospun nanofibers
 - D) None from the list
15. What is a major challenge in drug delivery within engineered tissues?
 - A) Avoiding polymer degradation
 - B) Preventing cell adhesion
 - C) Eliminating the need for ECM interactions
 - D) Achieving controlled and site-specific release
16. Which technique is commonly used to incorporate growth factors into polymeric scaffolds?
 - A) Chemical crosslinking
 - B) Electrospinning
 - C) Surface etching
 - D) Encapsulation
17. What is the role of ECM in polymer-based drug delivery?
 - A) ECM provides a natural environment that enhances cellular response and drug activity
 - B) ECM acts as a barrier to drug diffusion
 - C) ECM prevents scaffold degradation
 - D) ECM inhibits cell attachment to scaffolds
18. Which tissue type has been successfully engineered using bioreactors for clinical applications?
 - A) Cartilage
 - B) Skin
 - C) Bone
 - D) All from the list
19. Which bioreactor feature is crucial for musculoskeletal tissue engineering (tendon, ligament, muscle)?
 - A) Complete immobilization of the scaffold
 - B) Lack of cell interaction with the scaffold
 - C) Application of mechanical loading
 - D) The use of static culture conditions
20. What is a significant limitation of current bioreactor technologies?
 - A) High efficiency in tissue formation
 - B) The ability to eliminate the need for scaffolds
 - C) Difficulty in replicating the complexity of native tissue environments
 - D) Lack of cellular response to bioreactor conditions

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Compare natural and synthetic polymers for biomedical applications. Discuss their advantages and challenges in scaffold fabrication.	8	L1	1	1,5,7
	b) Explain the role of polymeric scaffolds in tissue engineering. What properties make a scaffold suitable for cell growth and differentiation?	8	L1	2	1,5,7
2.	a) Describe the significance of cell migration in wound healing and tissue regeneration. What are the main stages of cell migration?	8	L1	1	1,5,7
	b) Highlight the common surface modification techniques for improving polymer biocompatibility? How are these modifications characterized?	8	L1	2	1,5,7
3.	a) With respect to cadherins, integrins, and selectins, discuss their structure, function, and role in tissue organization.	8	L1	1	1,5,7

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| b) | Give an account on the various types of cell junctions. | 8 | L1 | 2 | 1,5,7 |
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Unit – II

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| 4. | a) | Describe different strategies used for the delivery of growth factors in tissue engineering and regenerative medicine. How do ECM components contribute to controlled release mechanisms? | 8 | L1 | 4 | 3,5,7 |
| | b) | Discuss in detail about Drug delivery in engineered tissue. | 8 | L1 | 3 | 3,5,6 |
| 5. | a) | What are the key challenges faced in using ECM for tissue engineering applications? Discuss future trends and advancements in ECM-based regenerative medicine. | 8 | L1 | 3 | 3,5,7 |
| | b) | Describe polymer hydrogel-based drug delivery systems. How do they achieve controlled and sustained release of therapeutic agents? Provide relevant examples. | 8 | L1 | 4 | 3,5,6 |
| 6. | a) | Explain the concept of gradient structures in tissue engineering. How do these structures mimic natural tissue transitions, and what are their applications? | 8 | L1 | 3 | 3,5,7 |
| | b) | Give a general overview of Drug delivery. Add a note on various factors affecting drug delivery. | 8 | L1 | 4 | 3,5,6 |

Unit – III

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| 7. | a) | What are the critical parameters for optimizing cell seeding efficiency in bioreactors? Discuss techniques to enhance uniformity, cell viability, and scaffold integration. | 8 | L1 | 5 | 5,6,7 |
| | b) | What are the challenges of engineering neuronal tissues using bioreactors? Discuss the limitations of neural cell culture, scaffold design, and functional integration. | 8 | L1 | 5 | 5,6,7 |
| 8. | a) | Analyze the ethical and regulatory challenges in the clinical application of bioreactor-grown tissues. How do safety concerns, standardization, and approval processes affect the adoption of engineered tissues in medicine? | 8 | L1 | 5 | 5,6,7 |
| | b) | How does the integration of bioreactors with 3D bioprinting enhance tissue engineering outcomes? Discuss the advantages and limitations of combining these technologies for functional tissue development. | 8 | L1 | 5 | 5,6,7 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
